

SCIF2001 Research Roadmap

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1 Literature Review

Classical electromagnetism is able to successfully describe the wave phenomena of light, such as interference and diffraction, however it does not fully account for the statistics of photodetection. Einstein's paper introduced the idea of the quantisation of light, where light exchanges energy in discrete amounts, described by $E = h\nu$ (Einstein 1905), while Glauber (1963) developed a quantum theory of optical coherence based on the correlation functions. This provided a framework for distinguishing optical fields through their detection statistics rather than only their average intensity.

The second-order correlation function $g^{(a)}(\tau)$ measures the relative probability of detecting two photons separated by a time delay τ . Thermal light exhibits bunching behaviour, whereas coherent light has approximately Poissonian statistics with $g^{(2)}(0) = 1$. A particularly strong indication of non-classical light is antibunching behaviour, $g^{(2)}(0) < 1$, because simultaneous detections are suppressed. This was observed in resonance fluorescence and argued that this result cannot be explained by a fully classical field model (Kimble, Dagenais, and Mandel 1977).

In a related demonstration, it was shown that using a beam splitter and a triggered single-photon source produced strong anticorrelation between the two output detectors (Grangier, Roger, and Aspect 1986). This is consistent with a single photon only being detected at only output. This particular experiment provides a practical measurement for demonstrating the quantum nature of light.

- operational signatures of nonclassical light
- experiments that verify those signatures
- entanglement and bell tests maybe

2 Skills Developed

2.1 Laboratory Skills

During the couple of weeks leading up to this literature review, I, along with another student, developed the skills necessary to work with optical fibres in a laboratory environment. This included fibre preparation, laser alignment, and characterising both the quality of the alignment, as well as some properties of the fibre itself.

The first stage of preparing a length of fibre is stripping off the outer-most protective layer, as well as the cladding. After cleaning the remaining cladding off of the fibre using acetone, we utilised a microscope connected to a computer in order to verify that all material was removed from around the core of the fibre. After this, we used a cleaver to make a clean, controlled break in the fibre, which we also checked under the microscope (see Figs. ??, ??). If we learnt one thing during our time working with fibres, it is that any sort of defect or nick or chip in the end of the fibre would significantly degrade the quality of the laser entering the fibre, which would in turn impact any experiment we may undertake. Whilst in this setup the light in even a bad cleave would be sufficient to do an experiment with, in the more advanced setups in the future, this loss would correspond to a significant proportion of the usable photons.

Once we had a good length of fibre, we were able to align the tip of the fibre to laser light going through a microscope objective. This stage was by far both the most important part, and the most tedious. This involved using adjustment screws in all of the x , y , and z axes, which allowed the tip of the laser to be

placed as close as possible to both the brightest region of laser light, as well as the correct distance from the microscope objective to ensure the light is focused. Initially, we ran into issues with the tip of the fibre being slightly bent due to storage on a reel. We were able to correct for this by having a shorter length of the fibre protruding from the holder, and moving the objective closer to account for this loss of range in the z -axis.

This laser alignment was aided by the use of light power meter. The end of the optical fibre not aligned with the laser was directed at a photosensor, which was read by the power meter to quantify exactly how much light, in terms of Watts, was being carried through the fibre. From this, we were able to see the difference in power whilst aligning, as can be seen in Figs. ?? and ??.

We were also able to utilise this power meter to characterise the numerical aperture (NA) of the fibre. This describes the range of the angles that the optical fibre can accept, and it depends on the refractive index of the fibre and the so-called "acceptance cone", the range of angles where laser is accepted into the fibre ("Numerical Aperture" 2026).

These skills developed in the Higher-Year Lab will allow us to design, setup, and maintain a optical setup that utilises optical fibre for the control of the laser used in our experiments. This will be of particular use, as the equipment we will be using will involve using laser contained in fibre for a large portion of the experiment.

2.2 Soft Skills

In addition to the skills learnt in the Higher-Year Lab, I was also able to develop more "soft skills" through the provided workshops. Specifically, I was able to improve my record-keeping, time management, and risk assessment and prevention.

3 Proposed Methodology

fucked if i know

4 Project Plan

In order to maintain a schedule that aligns with the outcomes and due dates of the course, I am attempting to adhere to the following plan to meet these outcomes and dates. This timeline takes into account personal events, study load for other courses, and a small amount of built-in leniency in the (likely) event of not meeting these targets.

Week 5: fuck around

Week 6: find out

Week 7: its so over

Week 8: were so back

Week 9: generational crash out

Week 10: generational lock in

Week 11: early graduation with 100000 degrees because im so smart (:

5 References

References

Einstein, A. (1905). "Über Einen Die Erzeugung Und Verwandlung Des Lichtes Betreffenden Heuristischen Gesichtspunkt". In: *Annalen der Physik* 322.6, pp. 132–148. ISSN: 1521-3889. DOI: [10.1002/andp.19053220607](https://doi.org/10.1002/andp.19053220607). (Visited on 10/08/2026).

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Kimble, H. J., M. Dagenais, and L. Mandel (Sept. 1977). “Photon Antibunching in Resonance Fluorescence”. In: *Physical Review Letters* 39.11, pp. 691–695. DOI: [10.1103/PhysRevLett.39.691](https://doi.org/10.1103/PhysRevLett.39.691). (Visited on 10/08/2026).

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